



HR ATTRITION ANALYTICS

Why 237 of 1,470 employees left — and where to act first

Problem → Approach → Insights → Recommendations → Business impact

The Problem

HR Leadership could not see who was leaving, or why

What was broken

- Attrition data sat in disconnected HR records.
- Every analysis was manual, slow, and different between reporting cycles.
- Attrition at 16.1% — above the 10–15% industry range — with no view of which segments drive it.
- Retention budget was spent reactively, not on highest-risk groups.

Questions to answer

Which departments and roles lose most? Do pay and age matter? Do overtime, travel and marital status predict exits?

Cost of doing nothing

237

employees lost out of 1,470 headcount

5.1 yrs

avg tenure of leavers — experienced people walking out

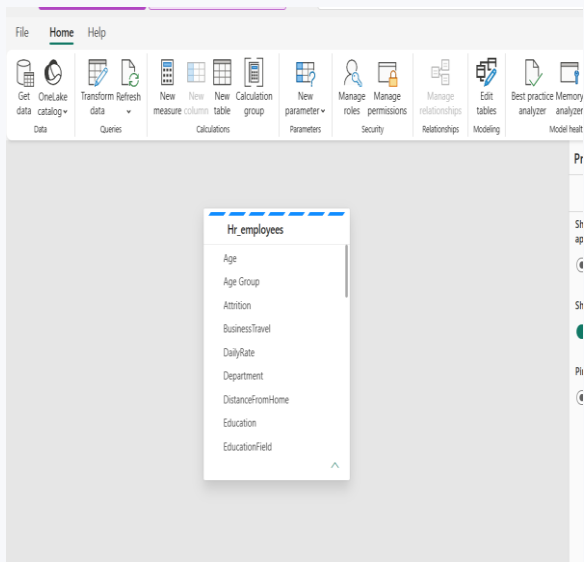
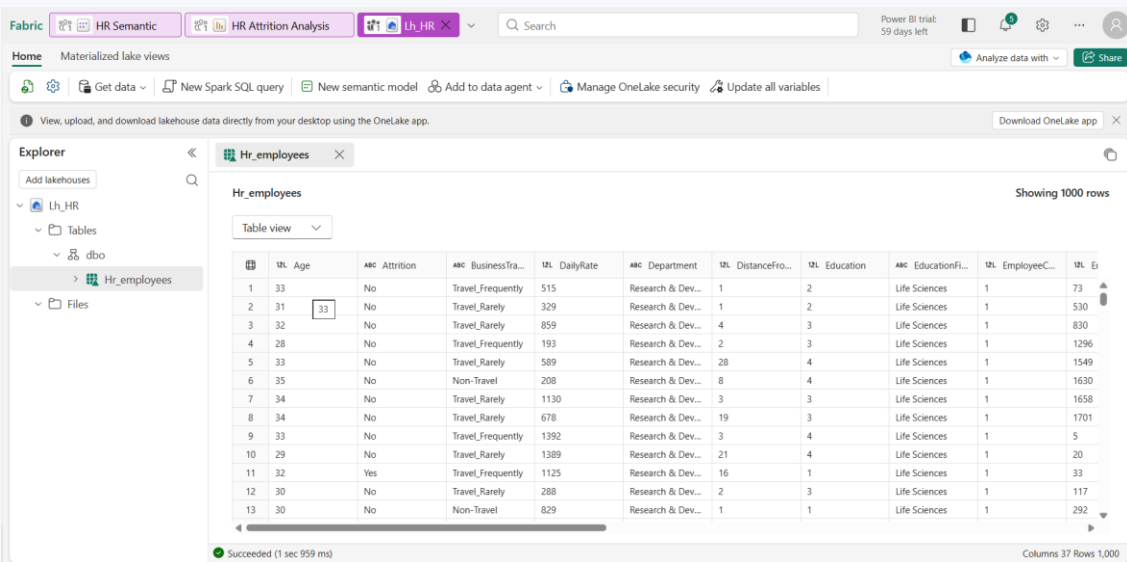
~9.2M

illustrative replacement cost*

*Assumption: replacement cost = 50% of annual pay (avg monthly income $6,503 \times 12 \times 0.5 \times 237$). Conservative; real range is often higher.

The Approach

End to end on Microsoft Fabric — raw CSV to governed dashboard

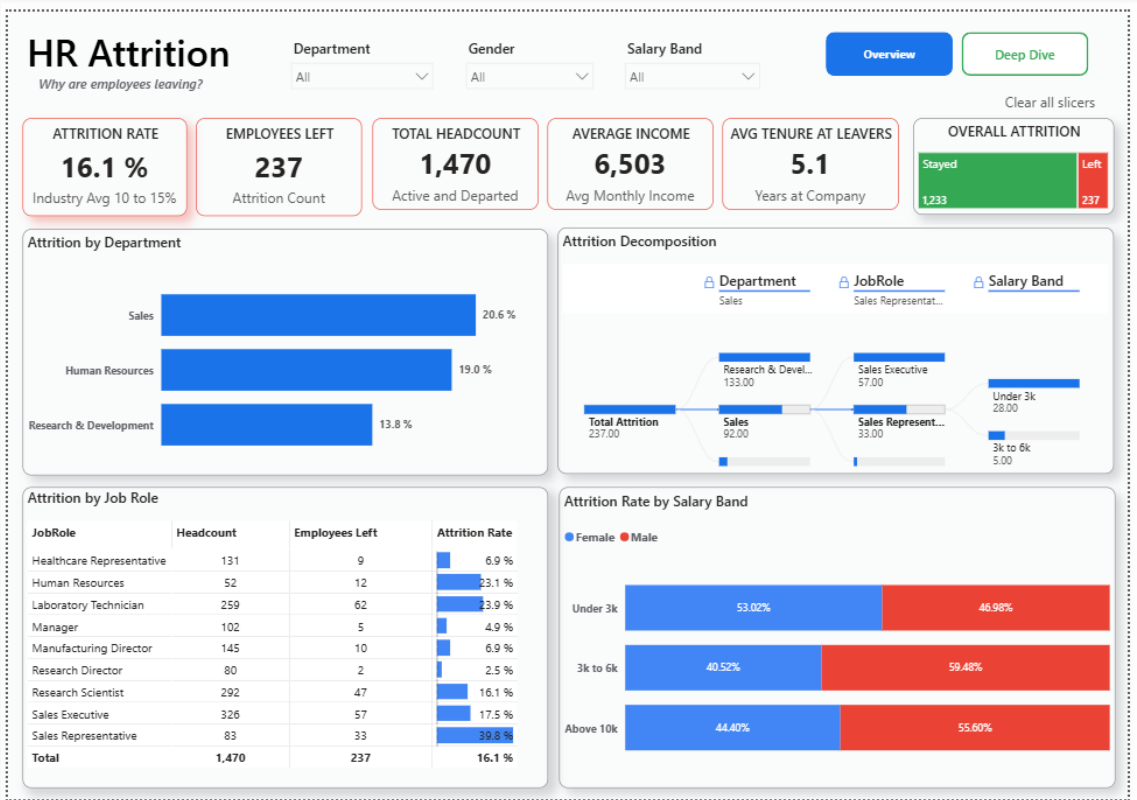


Why flat table?

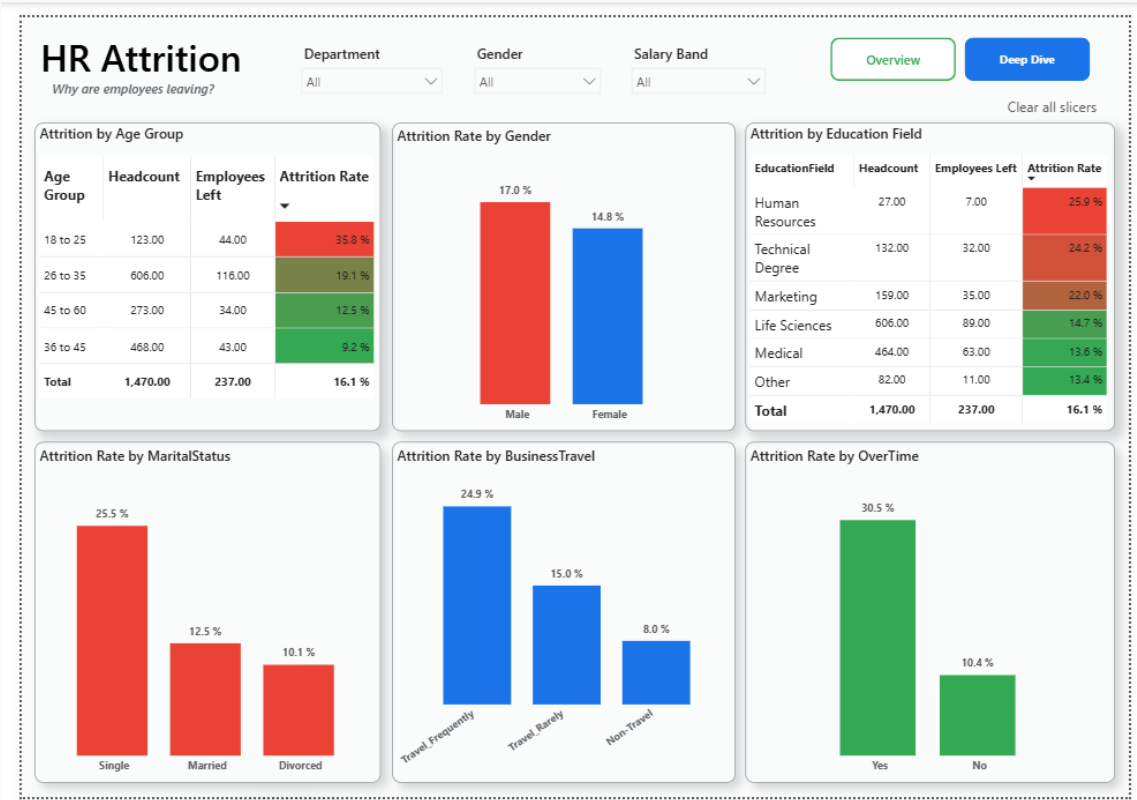
Under 2,000 rows, one row per employee. Star schema adds complexity, no benefit. Salary stays aggregate-only for general viewers.

The Dashboard

Two pages, cross-filterable: Overview and Deep Dive



Overview — KPIs, department split, decomposition tree, job role, salary band



Deep Dive — age, gender, education, marital status, travel, overtime

Key Insights

Attrition is concentrated, not spread evenly

39.8%

Sales Representatives

33 of 83 left — worst role

35.8%

Age 18–25

vs 9.2% for ages 36–45

30.5%

Overtime workers

~3x the 10.4% of non-overtime

24.9%

Frequent travelers

~3x the 8.0% of non-travelers

- Sales leads departments at 20.6%, ahead of HR (19.0%) and R&D (13.8%).
- Single employees leave 2x as often as married (25.5% vs 12.5%).
- Lab Technicians (23.9%) and HR roles (23.1%) are also above company average.
- Findings show correlation, not proof of cause — use as a guide for HR judgment.

Recommendations

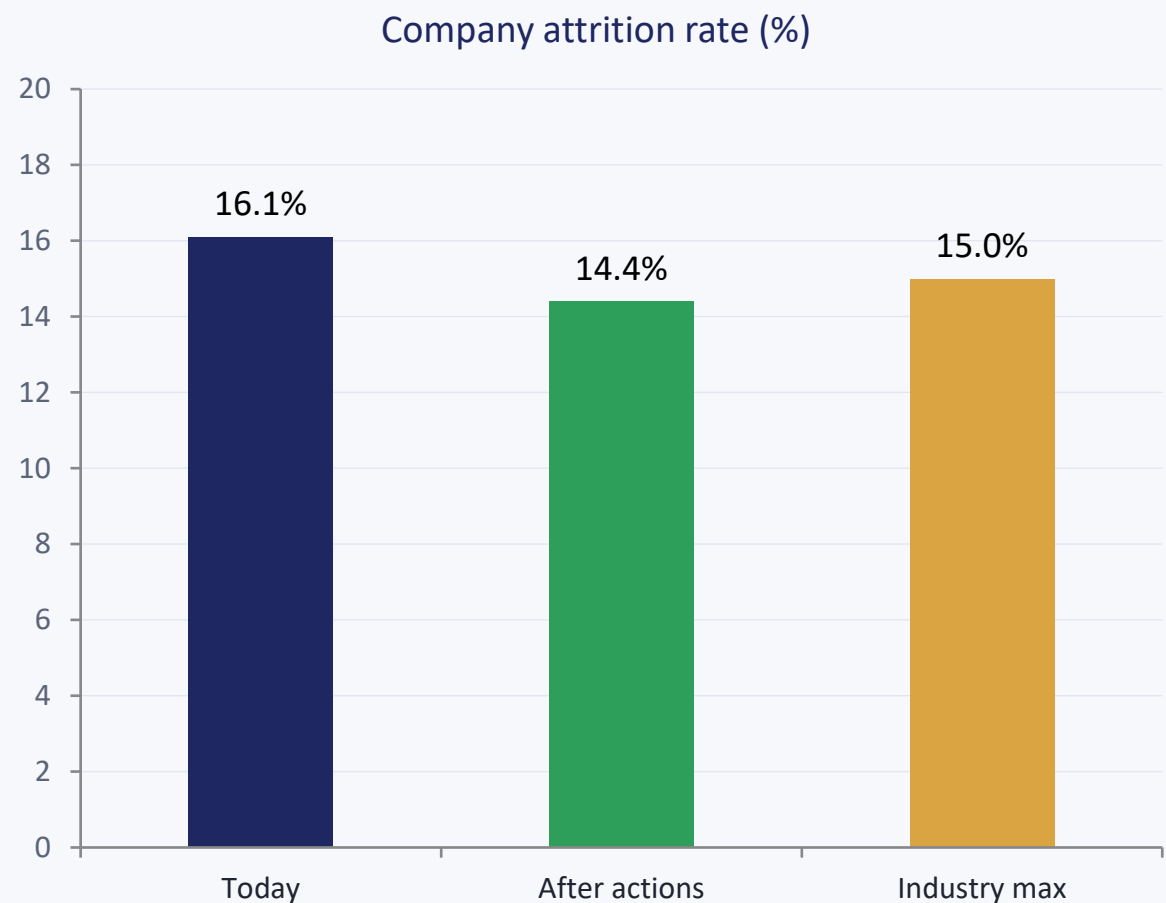
Four actions, ranked by evidence strength

#	Action	Target	Evidence
1	Fix Sales Rep pay and career path; review quota load	Sales Reps (83 people)	39.8% attrition; Under-3K band skews to leavers
2	Review overtime policy; rebalance workload	Overtime workers	30.5% vs 10.4% attrition
3	Structured onboarding + mentoring for new joiners	Age 18–25	35.8% attrition, highest age group
4	Cap travel / offer hybrid travel schedule	Frequent travelers	24.9% vs 8.0% for non-travel

Start with 1 and 2: biggest gap vs company average, easiest to act on this quarter.

Business Impact

Scenario: cut Sales Rep attrition to 25% and age 18–25 to 25%



~26

employees retained

Sales Reps: $83 \times (39.8\% \rightarrow 25\%) \approx 12$; Age 18–25: $123 \times (35.8\% \rightarrow 25\%) \approx 13$

–1.7 pts

attrition rate

$16.1\% \rightarrow \sim 14.4\%$, back inside the 10–15% industry range

~1.0M

replacement cost avoided

$26 \times 39,018$ (50% of avg annual pay) — illustrative

What This Project Shows

Skills delivered, limits, and next steps

Skills used

- Microsoft Fabric: Dataflow Gen2, Lakehouse
- Power BI semantic model, DAX
- Column-level security (governance)
- Business problem framing (BRD)
- Insight → recommendation → impact

Limits

- Point-in-time snapshot, no exit dates, so no trend
- Correlation only, not causation
- Cost figures are assumption-based
- Pay shown by band, not by individual

Next steps

- Add exit-date field for time-series trend
- ML risk scoring for active employees
- Fabric Data Activator alerts on thresholds
- Validate actions with a pilot in Sales